BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: BN-IX-XXII-I

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A (Version-C)

Time Allowed: 20 Min | **Maximum Marks:** 12 | **Paper Code:** BN-IX-XXII-I

Note: There are THREE sections in this paper i.e. A, B & C. Attempt section A and return it to the superintendent within the given time. You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. In groups from top to bottom ionization energy.

(A) Increase
(B) Decreases
(C) Unchanged
(D) Remains constant

Q2. Elements in a……have similar chemical properties.

(A) Period
(B) Group
(C) Both A and B
(D) Neither a nor b

Q3. C_2Cl_2 contains.

(A) Ionic bond
(B) Covalent bond
(C) Metallic bond
(D) Coordinate covalent bond

Q4. The study of hydrocarbons is called:

(A) Organic chemistry
(B) Inorganic chemistry
(C) Physical chemistry
(D) Nuclear Chemistry

Q5. The molecular mass of ethane (C_2H_6) is:

(A) 8 amu
(B) 18 amu
(C) 30 amu
(D) 44 amu

Q6. P subshell can accommodate a maximum of .....electrons

(A) 2 (B) 6

(C) 10 (D) 14

Q7. The SI unit of pressure is:

(A) NM^{-2} (B) N/m

(C) N/m^2 (D) Both A & C

Q8. How many electrons are shared in a single covalent bond.

(A) 2 (B) 4

(C) 6 (D) 8

Q9. the number of moles of solute dissolved per liter of solution is called:

(A) Percentage composition (B) Molarity

(C) Molality (D) All of these

Q10. NaOH is called.....electrolyte.

(A) Weak (B) Poor

(C) Non (D) Strong

Q11. Halogens react with non-metals forming.

(A) Ionic halides (B) Covalent Halides

(C) Metallic hydrides (D) Nitrides

Q12. Graphite is:

(A) Conductor (B) Superconductor

(C) Non-conductor (D) Insulator

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 32 | **Paper Code:** BN-IX-XXII-I

> Note: Attempt section B & C accordingly.

Q2. (Marks 32) Q.No 2. Attempt any EIGHT parts of the following. All parts carry equal marks.

i. How many elements are there in each of the following: (a) HF and Hf (b) $PoCl_2$ and $POCl_3$

ii. How many moles of hydrogen are there in $8.9x10^{23}$ hydrogen atom?

iii. Define isotopes with example of hydrogen?

iv. Why the s-block elements have two groups only?

v. Differentiate between polar and non-polar covalent bond?

vi. Why the densities of gases are lower than that of a liquid?

vii. Describe molarity?

viii. What is a redox-reaction? Give an example.

ix. Identify at least two groups which contain only metallic elements.

x. Differentiate between evaporation and boiling point.

xi. What is the difference between gram molecular mass and gram formula mass?

SECTION – C

Maximum Marks: 21 | **Paper Code:** BN-IX-XXII-I

> Note: Attempt any THREE of the following. All questions carry equal marks

Q3.

a) Calculate the molecular mass / formula mass of the following compounds. (i) C_6H_6 (ii) $AlCl_3$ (iii) Fe_2O_3 (iv) C_2H_6

b) What is the atomic number of an element? How does it differ from mass number?

Q4.

a) Explain the trends of atomic radius in the periodic table.

b) Write any four properties of ionic compounds?

Q5. Q.No:5. Define Charles's law and verify it graphically and diagrammatically?

b) Calculate the molarity of a solution if 30 cm^3 fo 2.5 M of $CaCO_3$ solution is diluted to 280 cm^3?

Q6.

a) Describe oxidizing and reducing agents?

b) Write down the reaction of chlorine with sodium hydroxide with balance equation?